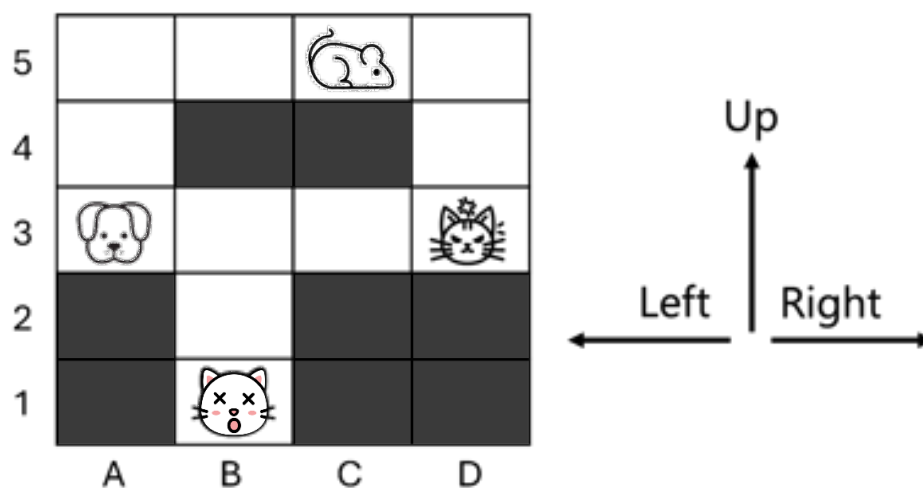


HOMEWORK SEARCH

Introduction to Artificial Intelligence

Question

A fat cat aims to beat a big boss mouse. The cat holds the map, which is shown below. The white squares are accessible while the black ones are not. For each step, the cat only moves to the next accessible square in one of the three directions: Up, Left, and Right. Each step uses one unit of energy.



Assume that the cat starts at square B1 and the big boss mouse is at square C5. The cat cannot directly beat the big boss mouse without picking up two friends along the way. Thus, the cat needs to first pick up its friends (at squares A3 and D3) and then heads to the big boss.

1. Compute the Manhattan distance ($dx + dy$) from accessible squares to the friends and his boss.

Square	A3	A4	A5	B1	B2	B3	B5	C3	C5	D3	D4	D5
Friend 1 (A3)												
Friend 2 (D3)												
Big boss (C5)												

2. Draw a search strategy to achieve the goals specified in (1), using A* with Manhattan distance. Since there are three goals, your strategy should be represented

by one or several search trees, depending on how you define the heuristic functions with Manhattan distance. Specify a triple of (f, g, h) - g is the path cost, h is the heuristic value, and $f = g + h$ - for each node on the search tree and mark the best path found.

Submission Requirements

Write your answers in the following template.

- Student ID: [Your Student ID]
- Name: [Your Name]
- Answers:
 1. [Your answer]
 2. [Your answer]

The submission document should be in PDF format and named as **StudentID.pdf**.